

40009 33

Haskell Macroprocessor

Submitters

kw1425

Emarking

```
1: Final Tests: Summary for kw1425 of c1
2: PPT 21
3: -----
4:
5:   Public Tests:
6:     student-tests/macro-test/macro/lookUp:          4 / 4
7:     student-tests/macro-test/macro/splitText:        4 / 4
8:     student-tests/macro-test/macro/combine:          4 / 4
9:     student-tests/macro-test/macro/getKeywordDefs:   9 / 9
10:    student-tests/macro-test/macro/replaceWords:     1 / 1
11:    student-tests/macro-test/macro/expand:           4 / 4
12:    student-tests/macro-golden/macro:                4 / 4
13:    original-tests/macro-test/macro/lookUp:          4 / 4
14:    original-tests/macro-test/macro/splitText:        4 / 4
15:    original-tests/macro-test/macro/combine:          4 / 4
16:    original-tests/macro-test/macro/getKeywordDefs:   9 / 9
17:    original-tests/macro-test/macro/expand:           3 / 3
18:    original-tests/macro-golden/macro:               3 / 3
19:
20: Git Repo: git@gitlab.doc.ic.ac.uk:lab2526_autumn/haskellmacroprocessor_kw1425.git
21: Commit ID: d061c
22: Git Diff: https://gitlab.doc.ic.ac.uk/lab2526_autumn/haskellmacroprocessor_kw1425/-/compare/master..d061c?from_project_id=129763
```

```

Final Tests                                Tests.hs: 1/2                                Katie Wan(kw1425):c1:21

1: import Test.Tasty (defaultMain, TestTree, testGroup)
2: import Test.Tasty.HUnit (testCase, (@?=), Assertion, HasCallStack)
3:
4: import MP
5:
6: main :: IO ()
7: main = defaultMain tests
8:
9: tests :: TestTree
10: tests = testGroup "macro"
11:   [ testGroup "lookUp" (numberedTests lookUpTests)
12:   , testGroup "splitText" (numberedTests splitTextTests)
13:   , testGroup "combine" (numberedTests combineTests)
14:   , testGroup "getKeywordDefs" (numberedTests getKeywordDefsTests)
15:   , testGroup "replaceWords" (numberedTests replaceWordsTests)
16:   , testGroup "expand" (numberedTests expandTests)
17:   ]
18:
19: lookUpTests :: [Assertion]
20: lookUpTests = [ lookUp "A" [(("A", 8), ("B", 9), ("C", 5), ("A", 7))] --> [8, 7]
21:               , lookUp "a" ([] :: [(String, Int)]) --> []
22:               , lookUp "a" [(("a", 9))] --> [9]
23:               , lookUp "a" [(("b", 9))] --> []
24:               ]
25:
26: splitTextTests :: [Assertion]
27: splitTextTests = [ splitText " .," "A comma, then some words." --> (" , .",
["A", "comma", "", "then", "some", "words", ""])
28:                 , splitText "" "" --> ("", [""])
29:                 , splitText "." "A.B" --> (".",
["A", "B"])
30:                 , splitText " " " A" --> (" ", ["",
"A"])
31:                 ]
32:
33: combineTests :: [Assertion]
34: combineTests = [ combine " , ." ["A", "comma", "", "then", "some", "words", ""] -->
["A", " , ", "comma", "", "", " , ", "then", " , ", "some", " , ", "words", ". ", ""]
35:               , combine "" [""] -->
[""]
36:               , combine "." ["A", "B"] -->
["A", ". ", "B"]
37:               , combine " " ["", "A"] -->
["", " ", "A"]
38:               ]
39:
40: getKeywordDefsTests :: [Assertion]
41: getKeywordDefsTests = [ getKeywordDefs ["$rule Reproduce this precisely -- or
else!!"] --> [("$rule", "Reproduce this precisely -- or else!!")]
42:                     , getKeywordDefs ["$x Define x", "$y 55"]
--> [("$x", "Define x"), ("y", "55")]
43:                     , getKeywordDefs ["$a A", "$b B", "$c C"]
--> [("$a", "A"), ("b", "B"), ("c", "C")]
44:                     , getKeywordDefs []
--> []
45:                     , getKeywordDefs ["", "$a A", ""]
--> [("$a", "A")]
46:                     , getKeywordDefs ["$x-y-z $$$"]
--> [("$x-y-z", "$$$")]
47:                     , getKeywordDefs ["$$ something to think about"]
--> [("$", "something to think about")]
48:                     , getKeywordDefs ["$ meanie!"]
--> [("$", "meanie!")]
49:                     , getKeywordDefs ["$var Tristan Allwood"]
--> [("$var", "Tristan Allwood")]
50:                     ]

```

```

Final Tests                                Tests.hs: 2/2                                Katie Wan(kw1425):c1:21

51:
52: replaceWordsTests :: [Assertion]
53: replaceWordsTests = [ replaceWords [("$name", "katie")] ["Hello", " ", "$name",
"" ] --> "Hello katie"]
54:
55: expandTests :: [Assertion]
56: expandTests = [ expand "The capital of $1 is $2" "$1 Peru\n$2 Lima." --> "The
capital of Peru is Lima."
57:               , expand "The time is $a" "$a now." --> "The
time is now."
58:               , expand "Keywords (e.g. $x, $y, $z...) may appear anywhere, e.g.
<$here>."
59:               , expand "$x $a\n$y $b\n$z $c\n$here $this-is-one"
--> "Keywords (e.g. $a, $b, $c...) may appear anywhere, e.g.
<$this-is-one>."
61:               , expand "Welcome to $town, where $name was born in $birth-date."
"$name William Shakespeare\n$birth-date 1564\n$town Stratford upon Avon"
62:               --> "Welcome to Stratford upon Avon, where William
Shakespeare was born in 1564."
63:               ]
64:
65: -----
66: -- HELPERS
67:
68: {-|
69: This function ensures that its first argument is the same as the second one
70: -}
71: infix 1 -->
72: (==>) :: (Show a, Eq a, HasCallStack)
73:      => a -- ^ the actual value
74:      -> a -- ^ the expected value
75:      -> Assertion
76: (==>) = (@?=)
77:
78: {-|
79: This function just matches up a bunch of assertions to a numerical naming
system, allowing us to distinguish them.
80:
81: If we wanted, we could provide descriptions to them instead...
82: -}
83: numberedTests :: [Assertion] -> [TestTree]
84: numberedTests = zipWith (\n -> testCase ("#" ++ show n)) ([1..] :: [Integer])
85:

```

```

Final Tests                               Golden.hs: 1/3                               Katie Wan(kw1425):c1:21

1: import Test.Tasty (defaultMain, TestTree, testGroup)
2:
3: import Test.Tasty.Golden
4:
5: import MP (expand)
6:
7: import System.IO (withBinaryFile, IOMode(ReadMode, WriteMode), hPutStr,
hGetContents, Handle)
8: import System.Directory (createDirectoryIfMissing, findExecutable,
removeDirectory, listDirectory)
9: import Control.Monad (when)
10: import Control.Exception (catch, throwIO, SomeException)
11: import Control.DeepSeq (($!))
12:
13: -- you may wish to run this specifically with
14: -- cabal test macro-golden --test-options="--color always --delete-output onpass"
15: -- this will ensure that the output files are deleted if the tests pass
16: main :: IO ()
17: main = do
18:   -- this checks to see if the "diff" command is available on your system
19:   -- if it is, this makes for nicer test failure messages
20:   -- if not, oh well
21:   diff <- findExecutable "diff"
22:   withDirectory "test/out" (defaultMain (golden diff))
23:
24: {-
25: Golden Testing
26: =====
27:
28: This testing file is doing "golden testing". This relies on the existence of
so-called
29: "golden tests", which are usually files with a specific contents. The idea is
that
30: a test will perform something that writes to a file, and then the new file is
compared
31: for exact equality with the corresponding golden test file. If they match,
great! If not,
32: something is wrong.
33:
34: It's a interesting approach, because it usually testing an entire system and can
be quite
35: brittle. As an example, some compilers have golden tests for their outputted
36: assembly/bytecode: when the compiler changes so that it generates different
code, it
37: will fail the golden tests. The interesting thing is that this means one of two
things:
38: the behaviour of the compiler has _intentionally_ changed, in which case the
golden test
39: is updated; or there was a bug introduced, which means the golden test remains
unchanged.
40: When a golden test fails, then, it is up to the tester to manually check and
determine
41: what the situation is. In fact, in this system, the golden tests are often
generated
42: on-demand by the old compiler!
43:
44: In our case, we are just using golden testing as a convenient way of testing the
IO
45: of your system, as opposed to you having to manually run these tests and then
inspect
46: the output yourself to see if they are right. In this sense, the "golden test
generator"
47: is simply our model solution.
48: -}
49:
50: golden :: Maybe FilePath -> TestTree

```

```

Final Tests                               Golden.hs: 2/3                               Katie Wan(kw1425):c1:21

51: golden diff = testGroup "macro" [ shakespeareTest diff
52:   , hardyTest diff
53:   , poohTest diff
54:   -- this tests the behaviour of the extension,
you can uncomment
55:   -- if you wish!
56:   , extensionTest diff
57: ]
58:
59: shakespeareTest :: Maybe FilePath -> TestTree
60: shakespeareTest diff =
61:   goldenTest diff "welcome message for Shakespeare"
62:     "test/golden/welcomeShakespeare.golden"
63:     "templates/welcome.txt"
64:     "info/shakespeare.info"
65:     "test/out/welcomeShakespeare.txt"
66:
67: hardyTest :: Maybe FilePath -> TestTree
68: hardyTest diff =
69:   goldenTest diff "welcome message for Hardy"
70:     "test/golden/welcomeHardy.golden"
71:     "templates/welcome.txt"
72:     "info/hardy.info"
73:     "test/out/welcomeHardy.txt"
74:
75: poohTest :: Maybe FilePath -> TestTree
76: poohTest diff =
77:   goldenTest diff "project information for Winnie the Pooh"
78:     "test/golden/pooh.golden"
79:     "templates/poohTem.html"
80:     "info/pooh.info"
81:     "test/out/pooh.html"
82:
83: extensionTest :: Maybe FilePath -> TestTree
84: extensionTest diff =
85:   goldenTest diff "multiple template expansion with authors"
86:     "test/golden/welcomeAuthors.golden"
87:     "templates/welcome.txt"
88:     "info/authors.info"
89:     "test/out/welcomeAuthors.txt"
90:
91: -- Internal things ('ere be Dragons)
92: -----
93:
94: goldenTest :: Maybe FilePath -> String -> FilePath -> FilePath -> FilePath ->
FilePath -> TestTree
95: goldenTest diff name reference template info output =
96:   testType diff name reference output (testOn template info output)
97:   where
98:     -- this is where we can decide whether or not we should try and diff the
output
99:     testType Nothing = goldenVsFile
100:    testType (Just diff) = flip goldenVsFileDiff (\ref new -> [diff, "-u", ref,
new])
101:
102: -- All the file IO must be done in binary mode to avoid issues with golden
testing on Windows
103: testOn :: FilePath -> FilePath -> FilePath -> IO ()
104: testOn template info output = do
105:   t <- readFile template
106:   i <- readFile info
107:   writeBinaryFile output (expand t i)
108:   where
109:     writeBinaryFile path stuff = withBinaryFile path WriteMode (\file ->
hPutStr file stuff)
110:

```

```
111: removeDirectoryIfEmpty :: FilePath -> IO ()
112: removeDirectoryIfEmpty dir =
113:   do contents <- listDirectory dir
114:   when (null contents) (removeDirectory dir)
115:
116: {-|
117:   This function sets up a directory for use within the given IO action: if it is
118:   empty after the action has completed, it is deleted. It will always be deleted
119:   even if the action throws an exception
120: -}
121: withDirectory :: FilePath -> IO () -> IO ()
122: withDirectory dir action = do
123:   createDirectoryIfMissing True dir
124:   action `catch` \e -> do
125:     removeDirectoryIfEmpty dir
126:     throwIO (e :: SomeException)
127:
```

```

1: module MP (separators, lookUp, splitText, combine, getKeywordDefs, replaceWords,
expand) where
2:
3: type FileContents = String
4:
5: type Keyword      = String
6: type KeywordValue = String
7: type KeywordDefs  = [(Keyword, KeywordValue)]
8:
9: separators :: String
10: separators = " \n\t.,;!\\"' () <>/\\"
11:
12: -----
13:
14: {-|
15: This function will look up a key in a list of key-value pairs,
16: returning all the values that match with that key.
17:
18: > lookUp "A" [("A", 8), ("B", 9), ("C", 5), ("A", 7)] == [8, 7]
19: -}
20: lookUp :: String -> [(String, a)] -> [a]
21: lookUp k p = [y | (x, y) <- p, x == k]
22:
23: -- recursion
24: -- lookUp _ [] = []
25: -- lookUp k ((x, y):xys)
26: --   | x == k = y : lookUp k xys
27: --   | otherwise = lookUp k xys
28:
29: -- tail recursion
30: -- lookUp k xs = go xs []
31: --   where
32: --     go :: [(String, a)] -> [a] -> [a]
33: --     go [] acc = reverse acc
34: --     go ((x, y):xys) acc
35: --       | x == k = go xys (y:acc)
36: --       | otherwise = go xys acc
37:
38: {-|
39: This function will break up a string with some given separator
40: characters, returning both the list of separators found between
41: each "word" and the words themselves.
42: -}
43: splitText :: [Char] -- ^ the separators to split on
44:           -> String -- ^ the string to split
45:           -> ([Char], [String])
46: splitText _ "" = ([], [])
47: splitText x (y:ys)
48:   | y `elem` x = (y:xs, "" : yss)
49:   | otherwise = (xs, zss)
50:   where
51:     (xs, yss) = splitText x ys
52:     (hs:tss) = yss
53:     zss = ((y:hs):tss)
54:
55: {-|
56: This function interleaves the characters from the first argument
57: list with the strings in the second argument. The second list must
58: be non-empty.
59: -}
60: combine :: [Char] -> [String] -> [String]
61: combine "" x = x
62: combine (x:xs) (y:ys) = y : [x] : combine xs ys
63:
64: {-|
65: This function takes a list of lines and splits each line to

```

14.5
20

2/2

3/4

how could fold be used?

2/4

how could fold be used?

```

66: extract a list of keyword-definition pairs.
67:
68: > getKeywordDefs ["$x Define x", "$y 55"] == [("$x", "Define x"), ("y", "55")]
69: -}
70: getKeywordDefs :: [String] -> KeywordDefs
71: getKeywordDefs k = [(w, concat (combine os ws)) | ((_,os), (w:ws)) <- split]
72:   where
73:     split = [splitText " " x | x <- k]
74:
75: -- replaceWord :: KeywordDefs -> Keyword -> KeywordValue
76: -- replaceWord word@(h:_)
77: --   | h == '$' = w
78: --   | otherwise = word
79: --   where (w:_) = lookUp word key
80:
81: replaceWords :: KeywordDefs -> [String] -> String
82: replaceWords key x = concatMap replaceWord x
83:   where
84:     replaceWord :: String -> String
85:     replaceWord "" = ""
86:     replaceWord word@(h:_)
87:       | h == '$' = w
88:       | otherwise = word
89:       where (w:_) = lookUp word key
90:
91:
92: {-|
93: This function takes the contents of two files, one containing
94: a template and the other the definitions for each keyword
95: found within the template. It extracts the keyword-definition
96: information from the info file and uses it to expand the occurrences
97: of these keywords in the template file, producing new file contents
98: as a result.
99:
100: > expand "The capital $1 is $2" "$1 Peru\n$2 Lima." == "The capital of Peru is
Lima"
101: -}
102:
103: join :: String -> [String] -> String
104: join _ [] = []
105: join _ [x] = x
106: join sep (t:ts) = t ++ sep ++ join sep ts
107:
108:
109: expand :: FileContents -- ^ the template file contents
110:       -> FileContents -- ^ the info file contents
111:       -> FileContents
112: -- expand body key = replaceWords ks x
113: --   where
114: --     (_, k) = splitText "\n\r" key - lines
115: --     ks = getKeywordDefs k
116: --     x = uncurry combine (splitText separators body)
117: --     (ops, wds) = splitText separators body
118: --     x = combine ops wds
119:
120: -- extension
121: expand body key = join "-----\n" replacedWords
122:   where
123:     (_, keyList) = splitText "# " key
124:     k = map (snd.splitText "\n\r") keyList
125:     ks = map getKeywordDefs k
126:     x = uncurry combine (splitText separators body)
127:     replacedWords = [replaceWords keyDef x | keyDef <- ks]

```

3/3

use comments to explain complex functions

2/3

replaceWord word@('\$':_)
replaceWord w

this function is called intercalate, from Data.List

2/2

0.5 / 2

Final Tests

testResults.txt: 1/5

Katie Wan(kw1425):c1:21

```
1: ----- Test Output -----
2: copying macro.cabal from skeleton
3: copying cabal.project from skeleton
4: Config file path source is default config file.
5: Config file not found: /root/.config/cabal/config
6: Writing default configuration to /root/.config/cabal/config
7: Resolving dependencies...
8: Build profile: -w ghc-9.6.7 -O2
9: In order, the following will be built (use -v for more details):
10: - call-stack-0.4.0 (lib) (requires build)
11: - colour-2.3.6 (lib) (requires build)
12: - macro-0.1.0.0 (lib) (first run)
13: - os-string-2.0.8 (lib) (requires build)
14: - prettyprinter-1.7.1 (lib) (requires build)
15: - splitmix-0.1.3.1 (lib) (requires build)
16: - tagged-0.8.9 (lib) (requires build)
17: - unliftio-core-0.2.1.0 (lib) (requires build)
18: - ansi-terminal-types-1.1.3 (lib) (requires build)
19: - macro-0.1.0.0 (exe:macroprocessor) (first run)
20: - hashable-1.5.0.0 (lib) (requires build)
21: - random-1.3.1 (lib) (requires build)
22: - ansi-terminal-1.1.3 (lib) (requires build)
23: - async-2.2.5 (lib) (requires build)
24: - temporary-1.3 (lib) (requires build)
25: - QuickCheck-2.16.0.0 (lib) (requires build)
26: - prettyprinter-ansi-terminal-1.1.3 (lib) (requires build)
27: - typed-process-0.2.13.0 (lib) (requires build)
28: - optparse-applicative-0.19.0.0 (lib) (requires build)
29: - tasty-1.5.3 (lib) (requires build)
30: - tasty-quickcheck-0.11.1 (lib) (requires build)
31: - tasty-hunit-0.10.2 (lib) (requires build)
32: - tasty-golden-2.3.5 (lib) (requires build)
33: - macro-0.1.0.0 (test:macro-properties) (first run)
34: - macro-0.1.0.0 (test:macro-test) (first run)
35: - macro-0.1.0.0 (test:macro-golden) (first run)
36: Configuring library for macro-0.1.0.0...
37: Starting    call-stack-0.4.0 (lib)
38: Starting    colour-2.3.6 (lib)
39: Starting    os-string-2.0.8 (lib)
40: Building    colour-2.3.6 (lib)
41: Building    call-stack-0.4.0 (lib)
42: Building    os-string-2.0.8 (lib)
43: Preprocessing library for macro-0.1.0.0...
44: Building library for macro-0.1.0.0...
45: [1 of 1] Compiling MP          ( src/MP.hs, /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/opt/build/MP.o,
/tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/opt/build/MP.dyn_o )
46: Installing  call-stack-0.4.0 (lib)
47: Starting    prettyprinter-1.7.1 (lib)
48: Completed   call-stack-0.4.0 (lib)
49: Starting    splitmix-0.1.3.1 (lib)
50: Building    prettyprinter-1.7.1 (lib)
51: Building    splitmix-0.1.3.1 (lib)
52: Installing  colour-2.3.6 (lib)
53: Completed   colour-2.3.6 (lib)
54: Starting    tagged-0.8.9 (lib)
55: Installing  splitmix-0.1.3.1 (lib)
56: Installing  os-string-2.0.8 (lib)
57: Completed   splitmix-0.1.3.1 (lib)
58: Starting    unliftio-core-0.2.1.0 (lib)
59: Completed   os-string-2.0.8 (lib)
60: Configuring executable 'macroprocessor' for macro-0.1.0.0...
```

Final Tests**testResults.txt: 2/5****Katie Wan(kw1425):c1:21**

```
61: Building      tagged-0.8.9 (lib)
62: Building      unliftio-core-0.2.1.0 (lib)
63: Preprocessing executable 'macroprocessor' for macro-0.1.0.0...
64: Building executable 'macroprocessor' for macro-0.1.0.0...
65: [1 of 1] Compiling Main              ( app/Main.hs, /
/tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/x/macroprocessor/opt/build/macroprocessor/macroprocessor-tmp/Main.o )
66: [2 of 2] Linking /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/x/macroprocessor/opt/build/macroprocessor/macroprocessor
67: Installing      unliftio-core-0.2.1.0 (lib)
68: Completed      unliftio-core-0.2.1.0 (lib)
69: Starting        ansi-terminal-types-1.1.3 (lib)
70: Building        ansi-terminal-types-1.1.3 (lib)
71: Installing      tagged-0.8.9 (lib)
72: Starting        random-1.3.1 (lib)
73: Completed      tagged-0.8.9 (lib)
74: Starting        hashable-1.5.0.0 (lib)
75: Building        random-1.3.1 (lib)
76: Building        hashable-1.5.0.0 (lib)
77: Installing      ansi-terminal-types-1.1.3 (lib)
78: Completed      ansi-terminal-types-1.1.3 (lib)
79: Starting        ansi-terminal-1.1.3 (lib)
80: Building        ansi-terminal-1.1.3 (lib)
81: Installing      ansi-terminal-1.1.3 (lib)
82: Installing      prettyprinter-1.7.1 (lib)
83: Completed      ansi-terminal-1.1.3 (lib)
84: Completed      prettyprinter-1.7.1 (lib)
85: Starting        prettyprinter-ansi-terminal-1.1.3 (lib)
86: Building        prettyprinter-ansi-terminal-1.1.3 (lib)
87: Installing      hashable-1.5.0.0 (lib)
88: Completed      hashable-1.5.0.0 (lib)
89: Starting        async-2.2.5 (lib)
90: Building        async-2.2.5 (lib)
91: Installing      prettyprinter-ansi-terminal-1.1.3 (lib)
92: Completed      prettyprinter-ansi-terminal-1.1.3 (lib)
93: Starting        optparse-applicative-0.19.0.0 (lib)
94: Building        optparse-applicative-0.19.0.0 (lib)
95: Installing      async-2.2.5 (lib)
96: Completed      async-2.2.5 (lib)
97: Starting        typed-process-0.2.13.0 (lib)
98: Building        typed-process-0.2.13.0 (lib)
99: Installing      typed-process-0.2.13.0 (lib)
100: Completed      typed-process-0.2.13.0 (lib)
101: Installing      optparse-applicative-0.19.0.0 (lib)
102: Completed      optparse-applicative-0.19.0.0 (lib)
103: Starting        tasty-1.5.3 (lib)
104: Building        tasty-1.5.3 (lib)
105: Installing      random-1.3.1 (lib)
106: Completed      random-1.3.1 (lib)
107: Starting        temporary-1.3 (lib)
108: Starting        QuickCheck-2.16.0.0 (lib)
109: Building        temporary-1.3 (lib)
110: Building        QuickCheck-2.16.0.0 (lib)
111: Installing      temporary-1.3 (lib)
112: Completed      temporary-1.3 (lib)
113: Installing      tasty-1.5.3 (lib)
114: Completed      tasty-1.5.3 (lib)
115: Starting        tasty-hunit-0.10.2 (lib)
116: Starting        tasty-golden-2.3.5 (lib)
117: Building        tasty-hunit-0.10.2 (lib)
118: Building        tasty-golden-2.3.5 (lib)
119: Installing      tasty-hunit-0.10.2 (lib)
120: Completed      tasty-hunit-0.10.2 (lib)
```


Final Tests**testResults.txt: 3/5****Katie Wan(kw1425):c1:21**

```
121: Configuring test suite 'macro-test' for macro-0.1.0.0...
122: Installing tasty-golden-2.3.5 (lib)
123: Completed tasty-golden-2.3.5 (lib)
124: Configuring test suite 'macro-golden' for macro-0.1.0.0...
125: Preprocessing test suite 'macro-test' for macro-0.1.0.0...
126: Building test suite 'macro-test' for macro-0.1.0.0...
127: [1 of 1] Compiling Main ( test/Tests.hs, ✓
/tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-test/opt/build/macro-test/macro-test-tmp/Main.o )
128: Preprocessing test suite 'macro-golden' for macro-0.1.0.0...
129: Building test suite 'macro-golden' for macro-0.1.0.0...
130: [2 of 2] Linking /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-test/opt/build/macro-test/macro-test
131: [1 of 1] Compiling Main ( test/Golden.hs, ✓
/tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-golden/opt/build/macro-golden/macro-golden-tmp/Main.o )
132: [2 of 2] Linking /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-golden/opt/build/macro-golden/macro-golden
133: Installing QuickCheck-2.16.0.0 (lib)
134: Completed QuickCheck-2.16.0.0 (lib)
135: Starting tasty-quickcheck-0.11.1 (lib)
136: Building tasty-quickcheck-0.11.1 (lib)
137: Installing tasty-quickcheck-0.11.1 (lib)
138: Completed tasty-quickcheck-0.11.1 (lib)
139: Configuring test suite 'macro-properties' for macro-0.1.0.0...
140: Preprocessing test suite 'macro-properties' for macro-0.1.0.0...
141: Building test suite 'macro-properties' for macro-0.1.0.0...
142: [1 of 1] Compiling Main ( test/Props.hs, ✓
/tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-properties/opt/build/macro-properties/macro-properties-tmp/Main.o )
143: [2 of 2] Linking /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-properties/opt/build/macro-properties/macro-properties
144: Build profile: -w ghc-9.6.7 -O2
145: In order, the following will be built (use -v for more details):
146: - macro-0.1.0.0 (test:macro-test) (ephemeral targets)
147: Preprocessing test suite 'macro-test' for macro-0.1.0.0...
148: Building test suite 'macro-test' for macro-0.1.0.0...
149: Running 1 test suites...
150: Test suite macro-test: RUNNING...
151: macro
152:   lookUp
153:     #1: OK
154:     #2: OK
155:     #3: OK
156:     #4: OK
157:   splitText
158:     #1: OK
159:     #2: OK
160:     #3: OK
161:     #4: OK
162:   combine
163:     #1: OK
164:     #2: OK
165:     #3: OK
166:     #4: OK
167:   getKeywordDefs
168:     #1: OK
169:     #2: OK
170:     #3: OK
171:     #4: OK
172:     #5: OK
173:     #6: OK
174:     #7: OK
175:     #8: OK
176:     #9: OK
177:   replaceWords
178:     #1: OK
```

Final Tests**testResults.txt: 4/5****Katie Wan(kw1425):c1:21**

```
179: expand
180:   #1: OK
181:   #2: OK
182:   #3: OK
183:   #4: OK
184:
185: All 26 tests passed (0.01s)
186:
187: Test suite macro-test: PASS
188: Test suite logged to:
189: /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-test/opt/test/macro-0.1.0.0-macro-test.log
190: 1 of 1 test suites (1 of 1 test cases) passed.
191: Build profile: -w ghc-9.6.7 -O2
192: In order, the following will be built (use -v for more details):
193:  - macro-0.1.0.0 (test:macro-golden) (ephemeral targets)
194: Preprocessing test suite 'macro-golden' for macro-0.1.0.0...
195: Building test suite 'macro-golden' for macro-0.1.0.0...
196: Running 1 test suites...
197: Test suite macro-golden: RUNNING...
198: macro
199:   welcome message for Shakespeare:      OK (0.01s)
200:   welcome message for Hardy:            OK
201:   project information for Winnie the Pooh: OK
202:   multiple template expansion with authors: OK
203:
204: All 4 tests passed (0.02s)
205:
206: Test suite macro-golden: PASS
207: Test suite logged to:
208: /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-golden/opt/test/macro-0.1.0.0-macro-golden.log
209: 1 of 1 test suites (1 of 1 test cases) passed.
210: copying test from skeleton
211: copying templates from skeleton
212: copying info from skeleton
213: Build profile: -w ghc-9.6.7 -O2
214: In order, the following will be built (use -v for more details):
215:  - macro-0.1.0.0 (test:macro-golden) (file test/Golden.hs changed)
216:  - macro-0.1.0.0 (test:macro-test) (file test/Tests.hs changed)
217: Preprocessing test suite 'macro-golden' for macro-0.1.0.0...
218: Preprocessing test suite 'macro-test' for macro-0.1.0.0...
219: Building test suite 'macro-golden' for macro-0.1.0.0...
220: Building test suite 'macro-test' for macro-0.1.0.0...
221: [1 of 1] Compiling Main          ( test/Tests.hs, /
/tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-test/opt/build/macro-test/macro-test-tmp/Main.o ) [Source file changed]
222: [1 of 1] Compiling Main          ( test/Golden.hs, /
/tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-golden/opt/build/macro-golden/macro-golden-tmp/Main.o ) [Source file changed]
223: [2 of 2] Linking /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-test/opt/build/macro-test/macro-test [Objects changed]
224: [2 of 2] Linking /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-golden/opt/build/macro-golden/macro-golden [Objects changed]
225: Build profile: -w ghc-9.6.7 -O2
226: In order, the following will be built (use -v for more details):
227:  - macro-0.1.0.0 (test:macro-test) (ephemeral targets)
228: Preprocessing test suite 'macro-test' for macro-0.1.0.0...
229: Building test suite 'macro-test' for macro-0.1.0.0...
230: Running 1 test suites...
231: Test suite macro-test: RUNNING...
232: macro
233:   lookUp
234:   #1: OK
235:   #2: OK
236:   #3: OK
237:   #4: OK
```

Final Tests**testResults.txt: 5/5****Katie Wan(kw1425):c1:21**

```
238: splitText
239: #1: OK
240: #2: OK
241: #3: OK
242: #4: OK
243: combine
244: #1: OK
245: #2: OK
246: #3: OK
247: #4: OK
248: getKeywordDefs
249: #1: OK
250: #2: OK
251: #3: OK
252: #4: OK
253: #5: OK
254: #6: OK
255: #7: OK
256: #8: OK
257: #9: OK
258: expand
259: #1: OK
260: #2: OK
261: #3: OK
262:
263: All 24 tests passed (0.00s)
264:
265: Test suite macro-test: PASS
266: Test suite logged to:
267: /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-test/opt/test/macro-0.1.0.0-macro-test.log
268: 1 of 1 test suites (1 of 1 test cases) passed.
269: Build profile: -w ghc-9.6.7 -O2
270: In order, the following will be built (use -v for more details):
271: - macro-0.1.0.0 (test:macro-golden) (ephemeral targets)
272: Preprocessing test suite 'macro-golden' for macro-0.1.0.0...
273: Building test suite 'macro-golden' for macro-0.1.0.0...
274: Running 1 test suites...
275: Test suite macro-golden: RUNNING...
276: macro
277: welcome message for Shakespeare: OK
278: welcome message for Hardy: OK
279: project information for Winnie the Pooh: OK
280:
281: All 3 tests passed (0.01s)
282:
283: Test suite macro-golden: PASS
284: Test suite logged to:
285: /tmp/d20251015-34-a37417/dist-newstyle/build/x86_64-linux/ghc-9.6.7/macro-0.1.0.0/t/macro-golden/opt/test/macro-0.1.0.0-macro-golden.log
286: 1 of 1 test suites (1 of 1 test cases) passed.
287:
288: ----- Test Errors -----
289: Warning: The package list for 'hackage.haskell.org' is 22 days old.
290: Run 'cabal update' to get the latest list of available packages.
291:
292: test/Golden.hs:84:1: warning: [-Wunused-top-binds]
293:   Defined but not used: â\200\230extensionTestâ\200\231
294:
295: 84 | extensionTest diff =
296:    | ^^^^^^^^^^^^^^^
```